# 无穷小的运算

$o(x^n) \pm o(x^n) = o(x^n)$

$o(x^n) \pm o(x^k) = o(x^k)$ , $k<n$

如 $o(x^2) \pm o(x^3) = o(x^2)$

$m \cdot o(x^n) = o(x^n)$ , $m$为常数

$x^k \cdot o(x^n) = o(x^{n+k})$

$o(x^k) \cdot o(x^n) = o(x^{n+k})$

$\dfrac{o(x^n)}{x^k} = o(x^{n-k})$ , $k<n$

例：求极限 $\lim\limits_{x\to 0} \dfrac{\cos x - e^{-\frac{x^2}{2}}}{\sin^4 x}$

解：

$$原式 = \lim_{x\to 0} \frac{\left[1-\frac{1}{2!}x^2+\frac{1}{4!}x^4+o(x^4)\right] - \left[1+\left(-\frac{x^2}{2}\right)+\frac{\left(-\frac{x^2}{2}\right)^2}{2!}+o(x^4)\right]}{x^4}$$

$$= \lim_{x\to 0} \frac{-\frac{1}{12}x^4 + o(x^4)}{x^4}$$

$$= -\frac{1}{12}$$